

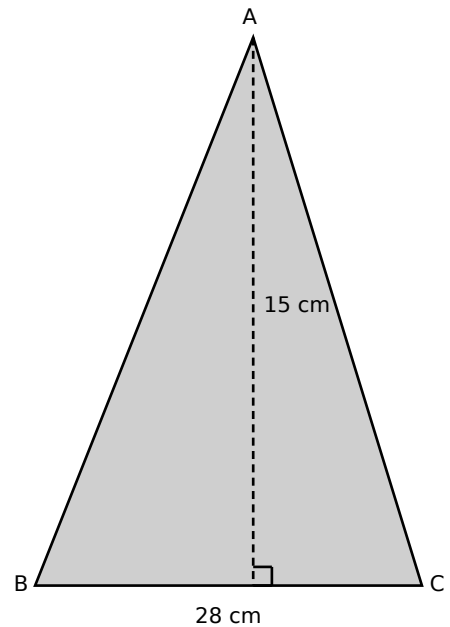
Set 2 - Shaded Areas

Level: Medium

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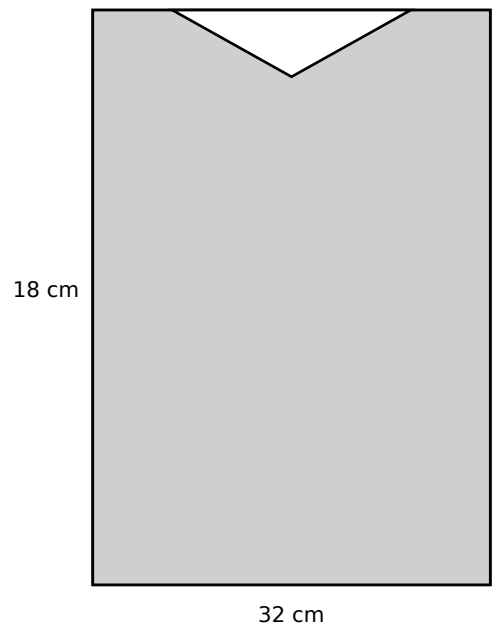
1. A shaded triangle has base 28 cm and perpendicular height 15 cm. Find its area.



Working:

Answer:

2. A rectangle measures 32 cm by 18 cm. An unshaded triangle at the top has base 20 cm and height 7 cm. Find the shaded area of the rectangle.



Working:

Answer:

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3. A rectangular paper is folded at one corner. The triangular folded part has base 6 cm and height 8 cm. Find the area of the folded part.

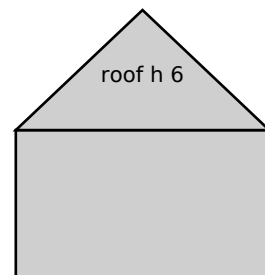


folded part: base 6, h 8

Working:

Answer:

4. A composite figure is made of a rectangle 16 cm by 12 cm and a triangular roof with base 16 cm and height 6 cm. Find the total area.



16 x 12

Working:

Answer:

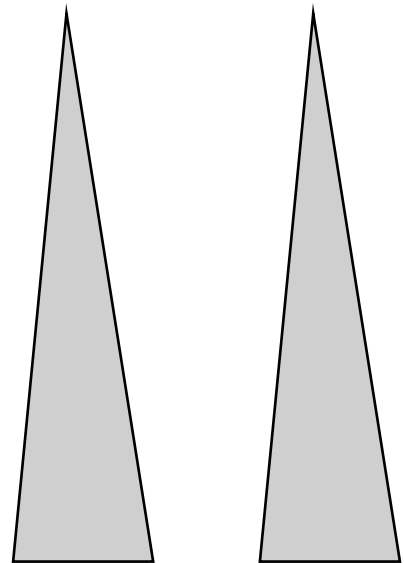
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5. A rectangle is split into 3 identical triangles. Each triangle has base 10 cm and height 12 cm. Two triangles are shaded. Find the shaded area.

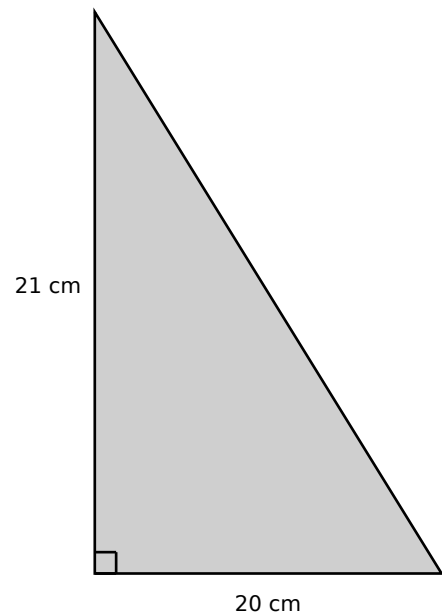


each: base 10 cm, height 12 cm

Working:

Answer:

6. A right-angled triangle has perpendicular sides 20 cm and 21 cm. Find its area.



Working:

Answer:

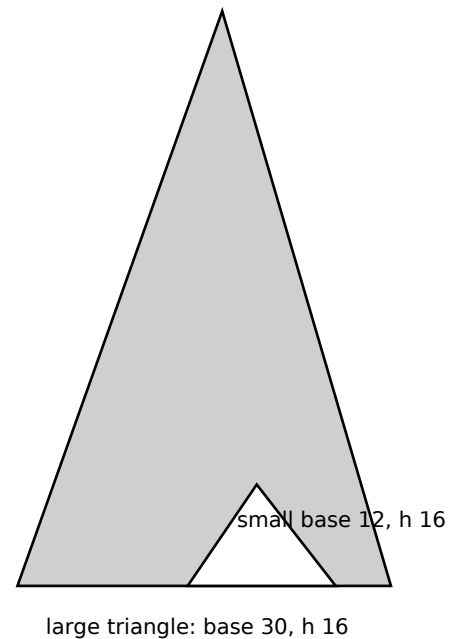
Set 2 - Shaded Areas

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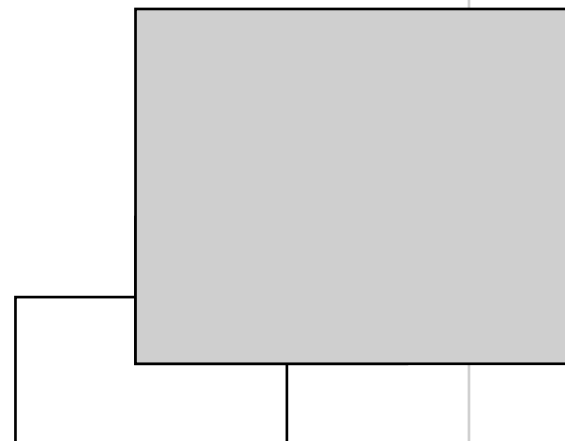
7. A large triangle has base 30 cm and height 16 cm. An unshaded triangle with the same height has base 12 cm. Find the shaded area.



Working:

Answer:

8. Two identical rectangles, each 16 cm by 7 cm, overlap. The overlapping area is 20 cm^2 . Find the area of the combined figure.



each 16 x 7; overlap 20

Working:

Answer:

Set 2 - Shaded Areas

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- 9.** Ben had some stickers. He gave $\frac{2}{5}$ of them to his brother and $\frac{1}{3}$ of the remainder to his friend. He had 36 stickers left. How many stickers did he have at first?

Working:

Answer:

- 10.** Cara used $\frac{3}{8}$ of a ribbon. She then used $\frac{1}{5}$ of the remainder. She had 50 cm of ribbon left. How long was the ribbon at first?

Working:

Answer:

Answer Sheet

Concise working is included for checking. Students may use other valid methods.

Qn	Answer	Short Working / Method
1	210 cm ²	$\frac{1}{2} \times 28 \times 15 = 210 \text{ cm}^2$
2	506 cm ²	Rectangle = 576; triangle = $\frac{1}{2} \times 20 \times 7 = 70$; shaded = 506 cm ²
3	24 cm ²	$\frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$
4	240 cm ²	Rectangle = 192; roof = $\frac{1}{2} \times 16 \times 6 = 48$; total = 240 cm ²
5	120 cm ²	Each = $\frac{1}{2} \times 10 \times 12 = 60$; 2 triangles = 120 cm ²
6	210 cm ²	$\frac{1}{2} \times 20 \times 21 = 210 \text{ cm}^2$
7	144 cm ²	Large = 240; unshaded = 96; shaded = 144 cm ²
8	204 cm ²	Total = $2 \times (16 \times 7) - 20 = 224 - 20 = 204 \text{ cm}^2$
9	90 stickers	Left = $\frac{3}{5} \times \frac{2}{3} = \frac{2}{5}$ of total. $\frac{2}{5} = 36$, so total = 90
10	100 cm	Left = $\frac{5}{8} \times \frac{4}{5} = \frac{1}{2}$ of total. $\frac{1}{2} = 50$, so total = 100 cm